

LINEAR AND CIRCULAR ARRANGEMENT

Directions for questions 1 to 4: These questions are based on the following information.

Four boys – Arun, David, Giri and Kiran and four girls – Prachi, Vaishali, Shivani and Malini attended a program and were seated in the first row which has eight chairs.

- (i) No two girls sit together. Each boy is adjacent to at least one girl.
- (ii) Prachi is adjacent to David. Vaishali is to the immediate right of Kiran.
- (iii) Arun and Giri are adjacent to each other.
- (iv) Shivani is not at any end and Vaishali is to the left of David but not adjacent to him.

The person who is sitting at the right end is

- ☐ a) Malini
- ☐ b) Prachi
- ☐ c) Vaishali
- ☐ d) David

Solution:

Given no two girls sit together, at least one boy must be in between each pair of girls.
But given Arun and Giri are adjacent, Girls must be seated at both ends.
Given Shivani is not at any end and Vaishali is to the right of Kiran and left of David, Prachi and Malini are at the ends.

As Prachi is adjacent to David who has Vaishali to his left but not adjacent, Shivani is adjacent to David.

M S D P

As Vaishali is to the immediate right of Kiran, he must be adjacent to Malini.

The final arrangement will be as follows.

M K V A/G G/A S D P

Prachi is at the right end.

Choice (B)

Which among the following pair of persons are definitely adjacent to each other?

- ☐ a) Malini and Arun
- ☐ b) Arun and Vaishali
- ☐ c) Shivani and Giri

- ☐ d) Prachi and David (Correct Answer - ✓)

How many persons are sitting between David and Kiran?

☐ a) 0

☐ b) 1

☐ c) 2

☐ d) 4 (Correct Answer - ✓)

Who is to the immediate left of Giri?

☐ a) Arun

☐ b) Vaishali

☐ c) Shivani

☐ d) Cannot be determined (Correct Answer - ✓)

Directions for questions 5 to 7: These questions are based on the following information.

Each of six persons A, B, C, D, E and F lives in a different floor of a six-storey building (The ground floor is named as the first floor, the floor above the ground floor is named as the second floor and so on)

A lives above E's floor. B lives below F's floor. At least two persons live between E's floor and D's floor. C does not live on an odd numbered floor, but lives below F's floor. B's floor is between F's floor and A's floor. Neither D nor A lives on an even numbered floor.

Who lives on the floor, which is immediately above A's floor?

☐ a) C

☐ b) B (Correct Answer - ✓)

☐ c) D

☐ d) F

Solution:

It is given that C does not live on odd number floor and lives below the floor on which F lives, so C lives on either second floor or fourth floor. It is given that A lives above the floor on which E lives, B lives below the floor on which F lives and B lives between the floors on which F and A live. So, F lives on either the fifth floor or the sixth floor.

So, we get the following possible cases.

Floor	Person			
	Case (1)	Case (2)	Case (3)	Case (4)
6	F		F	
5		F		F
4	C	C		
3				
2			C	C
1				

It is given that at least two persons live between the floors on which D and E live and D and A live on odd numbered floors. It is only possible in case (3). Thus the final arrangement is as follows.

Floor	Person
6	F
5	D
4	B
3	A
2	C
1	E

B lives immediately above A's floor.

Choice (B)

How many floors are there between E's floor and F's floor?

- ☐ a) One
- ☐ b) Two
- ☐ c) Three

☐ d) Four (Correct Answer - ✓)

Who lives immediately below C's floor?

☐ a) A

☐ b) D

☒ c) E (Correct Answer - ✓)

☐ d) B

Directions for questions 8 to 11: These questions are based on the following information.

Four couples are sitting in two rows, with four persons in each row, such that each person in one row is exactly opposite one person in the other row. Of them Arjun, Bharath, Chanti and Dheeraj are males and Anjali, Bharani, Chandra and Divya are females. We know the following additional information.

- (i) No female is either adjacent to or opposite her husband.
- (ii) The number of persons to the left of Dheeraj in his row is two.
- (iii) Anjali is the wife of Bharath.
- (iv) Bharani is sitting to the immediate right of Chanti and opposite Arjun.
- (v) Divya is to the immediate left of Chandra.
- (vi) Each female is sitting opposite a male.
- (vii) Arjun is sitting at the end of a row.

Who is the wife of Chanti?

☐ a) Anjali

☐ b) Bharani

☐ c) Chandra

☐ d) Divya

Solution:

From (iii) and (iv) Bharani is not the wife of Chanti or Arjun or Bharath.

∴ Bharani is the wife of Dheeraj.

∴ Bharani is not adjacent to Dheeraj.

From (ii), we have the following arrangement:

_____	<u>Dheeraj</u>	_____	_____
_____	_____	_____	_____

From (iv), Bharani cannot be to the immediate left of Dheeraj.

From (vi) and above results, the only possibility is that, Bharani is at the extreme right of her row which is opposite to Dheeraj's row as shown:

_____	<u>Dheeraj</u>	_____	<u>Arjun</u>
_____	_____	<u>Chanti</u>	<u>Bharani</u>

From (v), Divya is at the extreme left of her row and Bharath is opposite Divya.

The final arrangement is as follows:

<u>Bharath</u>	<u>Dheeraj</u>	<u>Anjali</u>	<u>Arjun</u>
<u>Divya</u>	<u>Chandra</u>	<u>Chanti</u>	<u>Bharani</u>

As Chandra and Chanti are adjacent to each other, Chandra is not the wife of Chanti.

∴ Chandra is the wife of Arjun.

⇒ Divya is the wife of Chanti.

Divya is the wife of Chanti.

Choice (D)

Who is sitting opposite Divya?

☐ a) Arjun

☒ b) Bharath (Correct Answer - ✓)

☐ c) Chanti

☐ d) Dheeraj

Who is sitting to the immediate right of Arjun?

☐ a) Anjali (Correct Answer - ✓)

☐ b) Bharath

☐ c) Chandra

☐ d) Divya

Who is the husband of Chandra?

☐ a) Dheeraj

☐ b) Bharath

☐ c) Arjun (Correct Answer - ✓)

☐ d) Chanti

Directions for questions 12 to 15: These questions are based on the following information.

A, B, C, D, E, F, G and H are sit around a circular table, some of them face the centre and the rest are face away from the centre. The following information is known about their seating.

B sits second to the left of G. F is an immediate neighbour of G and C. A sits adjacent to C and is facing D. D sits third to the right of C and second to the left of E, who is facing the centre. No three persons seated adjacent are facing the centre.

How many persons are sitting between H and F, when counted from the right of H?

☐ a) Three

☐ b) Two

☐ c) Four

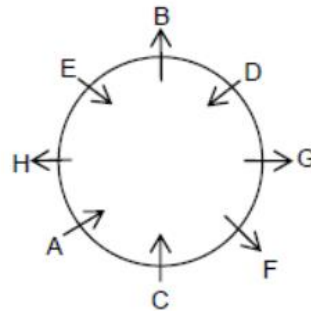
☐ d) One

Solution:

It is given that, B sits second to the left of G. F is an immediate neighbour of G and C. Hence, G either faces the centre or away from the centre. F sits to the immediate right of G and C sits second to the right of G. Given A sits adjacent to C and is facing D. Hence A and D are facing the centre. D sits third to the right of C and second to the left of E, who is facing the centre.

G is facing away from the centre. Since, if G faces the centre, we cannot place D to the second left of E, as E is facing the centre E sits adjacent to B and H sits adjacent to A and E. Given, no three persons seated adjacent are facing the centre. Hence B, H and F are facing away from the centre.

∴ The final arrangement is as shown below.



There are four persons sitting between H and F, when counted from the right side of H.
Choice (C)

Who sits second to the right of F?

- ☐ a) D
- ☐ b) C
- ☐ c) B

☐ d) A (Correct Answer - ✓)

Three of the following four are alike in a certain way based on the given information and so form a group. Find the one that does not belong to that group.

☐ a) B, E

☐ b) D, G

☒ c) G, F (Correct Answer - ✓)

☐ d) F, C

How many persons are facing away from the centre?

☐ a) Three

☒ b) Four (Correct Answer - ✓)

☐ c) Five

☐ d) Cannot be determined